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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026
SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCBCH100 - ESSENTIALS OF BIOCHEMISTRY: VITAMINS, HORMONES, ENZYMES AND NEUROTRANSMITTERS
(2024 Admission onwards)

Duration: 1 Hour 30 Minute **Maximum Marks:** 50

Part A – Multiple Choice Questions

Answer all questions (1 mark each)

1. Which coenzyme is derived from Pantothenic acid (Vitamin B5)?
   ANS: c) Coenzyme A (CoA) [U]/[CO1]

2. Find the reaction catalyzed by flavin adenine dinucleotide.
   ANS: a) Redox reaction [K]/[CO1]

3. Which of the following neurotransmitter is an inhibitory neurotransmitter?
   ANS: a) GABA [K]/[CO4]

4. Main function of oxidoreductase enzymes:
   ANS: c) Oxidation-reduction reactions [E]/[CO5]

5. Significance of enzyme specificity:
   ANS :d) Proper metabolic functioning [E]/[CO5]

Part B – Fill in the Blanks

(2 marks each)

6. Daily requirement of vitamin B12 during pregnancy and lactation: _____
   ANS: 2mg/day [K]/[CO1]

7. _____ is the plant form of vitamin D
   ANS: Ergocalciferol (Vitamin D_2) [K]/[CO1]

8. _____ is the father of endocrinology, and he discovered _____ disease
   ANS: Thomas Addison, Addison's Disease [K]/[CO2]

9. Antidiuretic hormone is related to the disorder _____
   ANS: Diabetes insipidus [U]/[CO3]

10. Temperature of maximum enzyme activity is called _____
ANS: Optimum Temperature [K]/[CO5]

Part C – Short Answer

(Answer any 5, 3 marks each)

11. What is xerophthalmia, and how is it related to Vitamin A deficiency?
ANS: Explain Xerophthalmia – 1.5 marks
Relation to Vitamin A deficiency- 1.5marks [K]/[CO1]

12. How is the secretion of oxytocin influencing social behavior and bonding?
ANS: Functions/anything of/about oxytocin -3marks
[U]/[CO4]

13. Describe the role of G-protein-coupled receptors (GPCRs) in peptide hormone signaling.
ANS: Explanation of GPCRS in signalling – 3marks [K]/[CO3]

14. List the various hormones produced by the gastrointestinal system.
ANS: Mention any three hormones (Gastrin, Secretin, cholecystokinin, ghrelin and leptin) – each carries one mark [K]/[CO3]

15. Why do heavy metals (e.g., lead, mercury) interfere with enzyme activity?
ANS: For attempt give full marks [E]/[CO5]

16. Define Michaelis constant (Km) and its significance.
ANS: Definition – 1mark
Significance -2marks [U]/[CO5]

17. What is the characteristic feature of oxidoreductases?
ANS: Function -1.5marks
Example 1.5marks [U]/[CO5]

Part D – Short Essay

(Answer any 4, 5 marks each)

18. Explain the functions and sources of Vitamin B complex vitamins.
ANS: Functions of Vitamin B Complex – 3marks [U]/[CO1]
Sources – 2mark

19. Describe the process of intracellular signalling involved in steroid hormone action.
ANS: Description – 3marks
Diagram- 2marks [K]/[CO3]

20. Describe the physiological significance of hormone transport mechanisms in maintaining homeostasis, considering the roles of carrier proteins and receptor types in hormone action.
ANS: Basic Concept – 1 mark
Carrier protein and significance -2marks
Role in Homeostasis- 2marks [U]/[CO2]

21. Compare and contrast the Lineweaver–Burk plot and Michaelis–Menten plot with graph and explain its significance.9

ANS: Linweaver Burkplot graph and explanation -3marks
Michaelis Menten equation and significance -2marks [E]/[CO5]

22. Describe what happens if there is too much enzyme but not enough substrate.

ANS: Explanation -3marks
Graph/Diagram – 2marks [U]/[CO5]

23. Analyze the Michaelis–Menten equation and explain the importance and significance of Km value.

ANS: Equation – 1mark
Importance and definition- 2 marks
Significance – 2marks [U]/[CO5]

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